13.2.5 Insights into availability zones

A region consists of multiple availability zones. Each availability zone consists of at least one isolated data center. The identifier for an availability zone consists of the identifier for the region (such as us-east-1) and a character (a, b, c, ...). So us-east-1a is the identifier for an availability zone in region us-east-1. To distribute resources across the different availability zones, the AZ identifier is mapped to one or multiple data centers randomly when creating an AWS account. This means us-east-1a might point to a different availability zone in your AWS account than it does in our AWS account.

We recommend that you take some time to explore the worldwide infrastructure provided by AWS. You can use the following commands to discover all regions available for your AWS account:

```
$ aws ec2 describe-regions                                    ①
{
  "Regions": [
    {
      "Endpoint": "ec2.eu-north-1.amazonaws.com",             ②
      "RegionName": "eu-north-1",                             ③
      "OptInStatus": "opt-in-not-required"                    ④
    },
    {
      "Endpoint": "ec2.ap-south-1.amazonaws.com",
      "RegionName": "ap-south-1",
      "OptInStatus": "opt-in-not-required"
    },
    [...]
    {
      "Endpoint": "ec2.us-west-2.amazonaws.com",
      "RegionName": "us-west-2",
      "OptInStatus": "opt-in-not-required"
    }
  ]
}
```

① Lists all regions available for your AWS account

② The endpoint URL, used to access the EC2 service in the region

③ The name of the region

④ Newer regions require an opt-in.

Next, to list all availability zones for a region, execute the following command and replace `$Region` with `RegionName` of a region from the previous output: